

Automatic Speech Recognition (ASR) Performance Across Individuals with and without Down Syndrome (DS)

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INTRODUCTION

ASR can result in:

- Increased feelings of competence and independence in individuals with disabilities [1]
- Improvements in speech and literacy [2,3]

Individuals with DS:

- Often struggle with communication due to hypotonia, hearing loss, and various communication disorders [4]
- Prefer ASR as an input-method [5]
- Could use ASR as an AAC intervention
- Are not often studied in connection with ASR

METHODS

- Used Sonix Speech-to-Text

Participants:

- 10 adults with DS (mean age=27.5, range: 18-39)
- 12 typical adults (mean age=18.66, range: 18-21)

Measures:

- Word Error Rate (WER):

$$WER = \frac{D+I+S}{N},$$

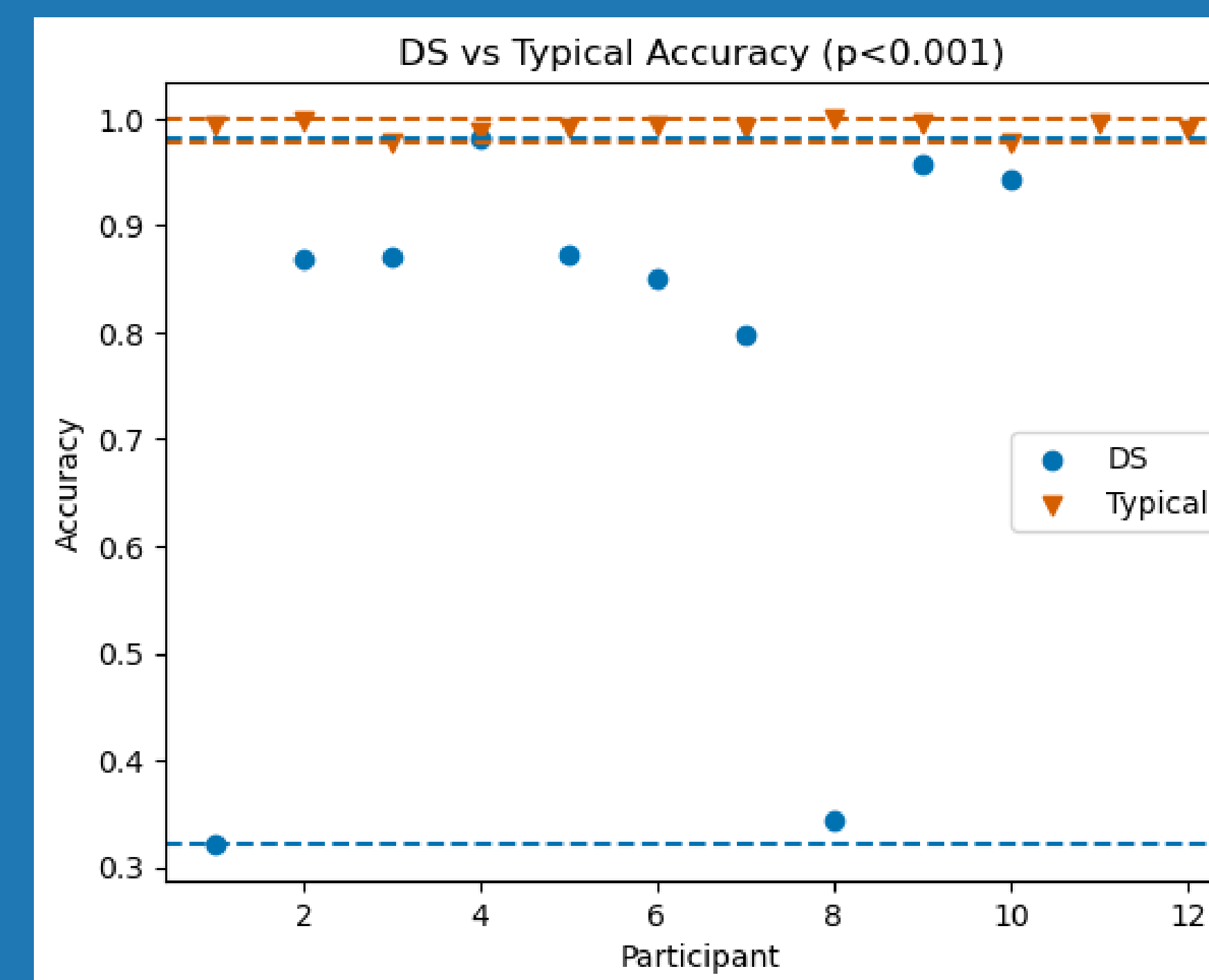
- Accuracy: $Acc = 1 - WER$
- Sounds matched in phonemic analysis using minimum Levenshtein distance

References:

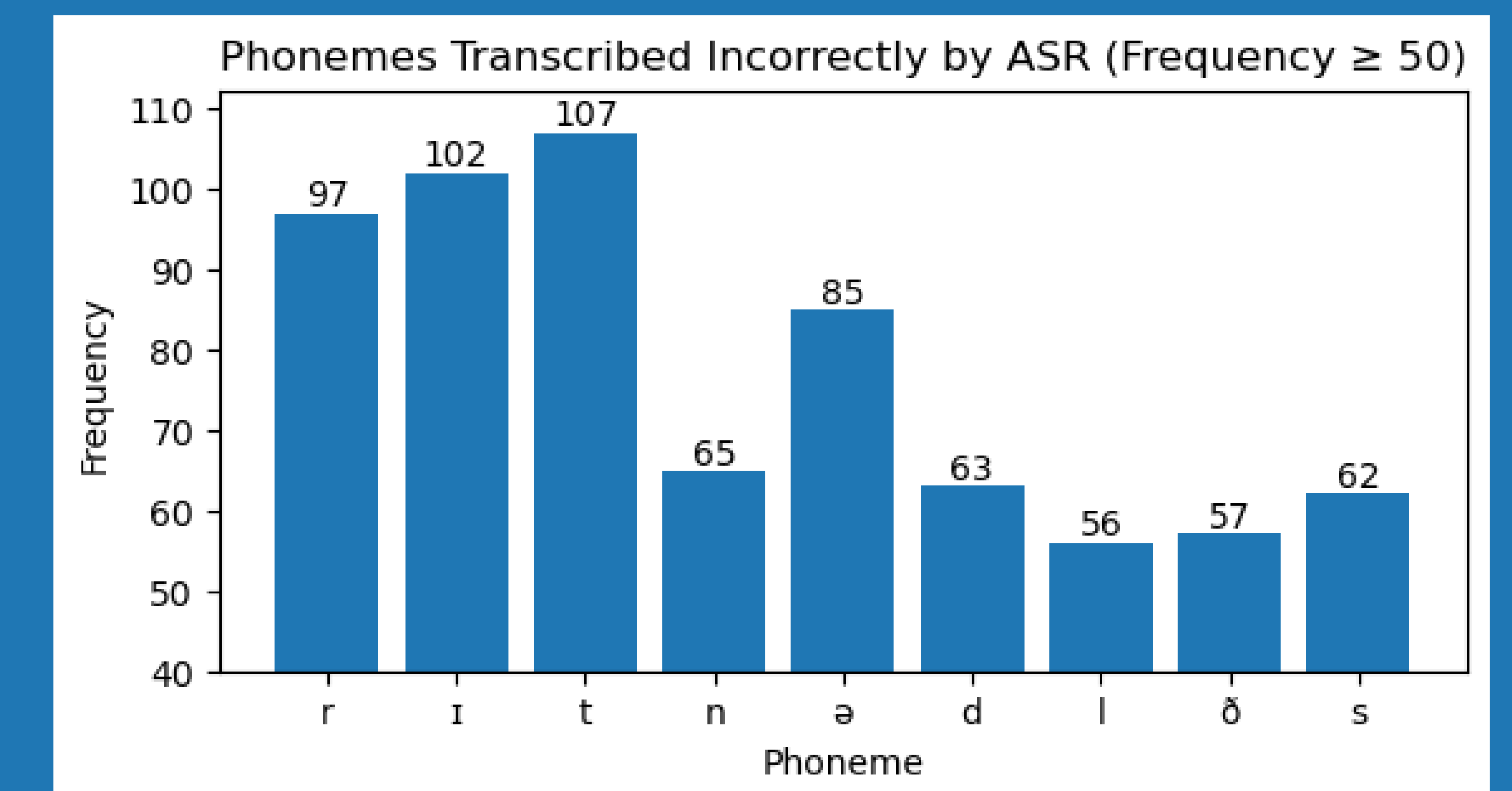
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- [5] Ruimin Hu, Jinjuan Feng, Jonathan Lazar, and Libby Kumin. 2013. Investigating input technologies for children and young adults with Down syndrome. *Univers. Access Inf. Soc.* 12, 1 (March 2013), 89-104.
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RESULTS

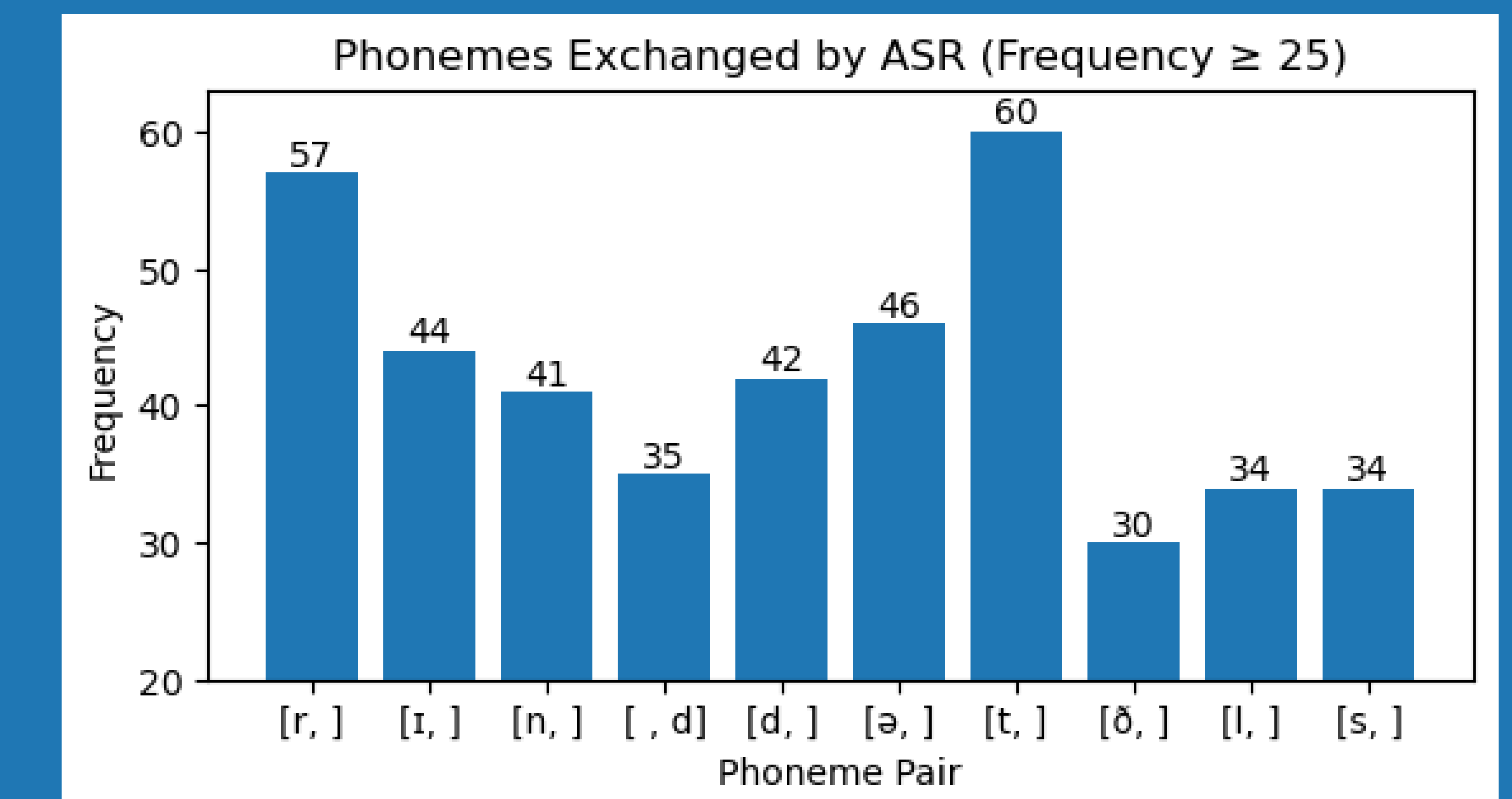
1. Sonix performed significantly worse for individuals with DS compared to typical individuals



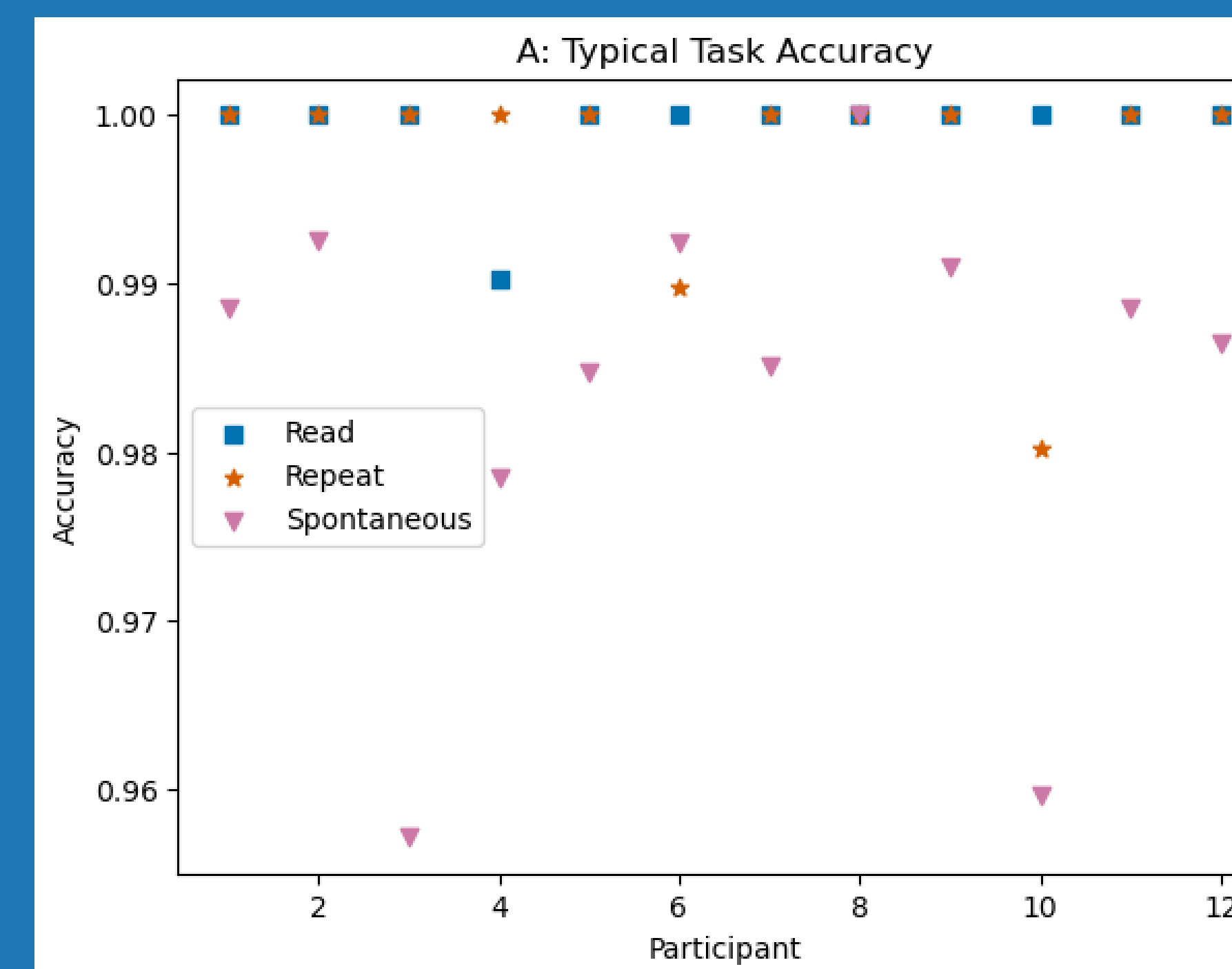
3a. Frequently misarticulated phonemes in DS are often transcribed incorrectly by Sonix



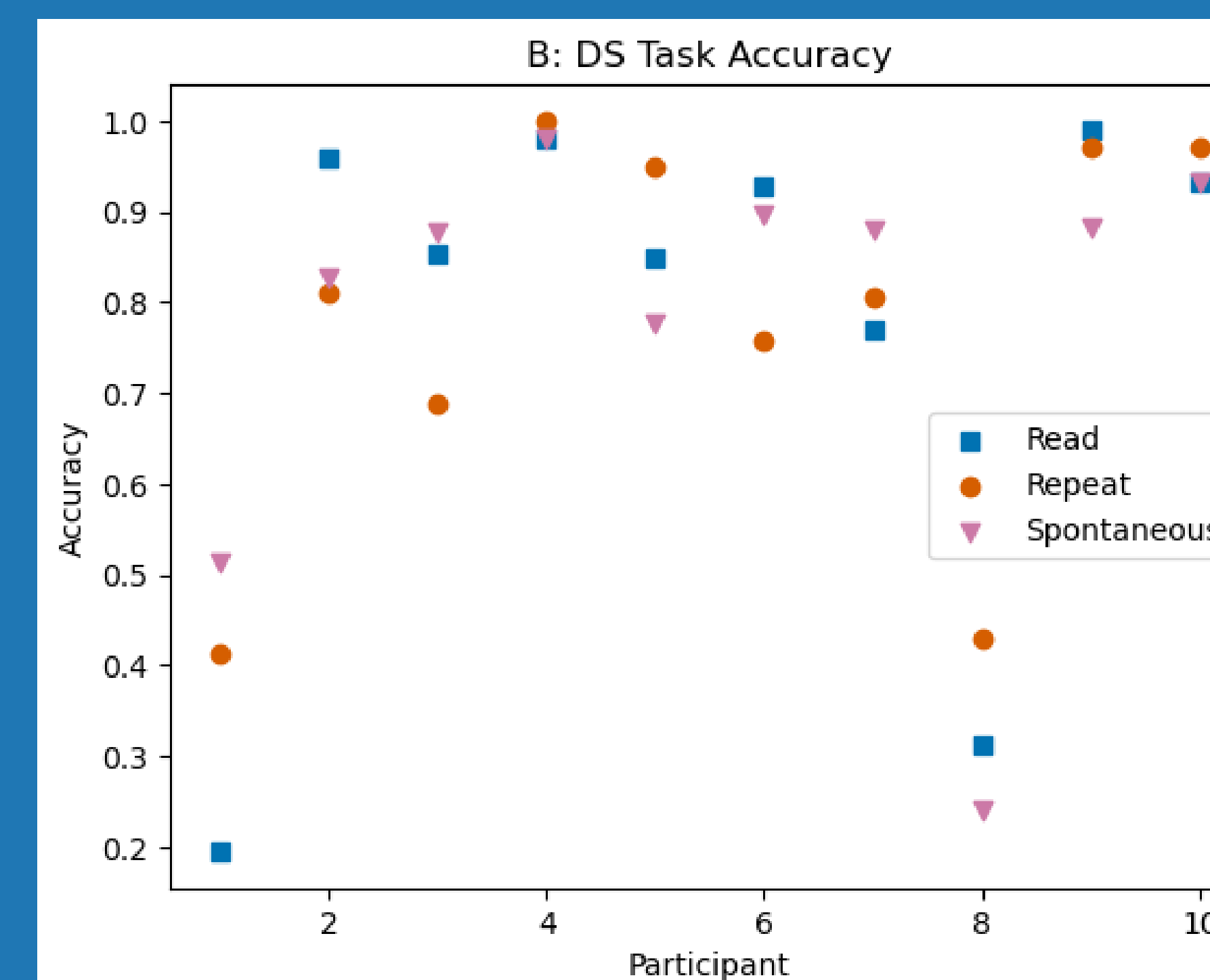
3b. Phonemes were mainly incorrectly deleted.



2a. Sonix had higher accuracy for read and repeat tasks in the typical group (p<0.01)



2b. No difference in task accuracy was found in the group with DS



DISCUSSION

- Sonix demonstrated higher accuracy for individuals with DS compared to IBM, Google, and Microsoft ASR models as reported by [6].
- Results in 2b may be due to lower literacy and poor verbal short-term memory in individuals with DS.
- Deletions in 3b may result from Sonix misclassifying speech as noise or may reflect cluster simplification in DS.
- By focusing on training with data including commonly misarticulated phonemes, an ASR model could be adapted for individuals with DS using less data and training time.